

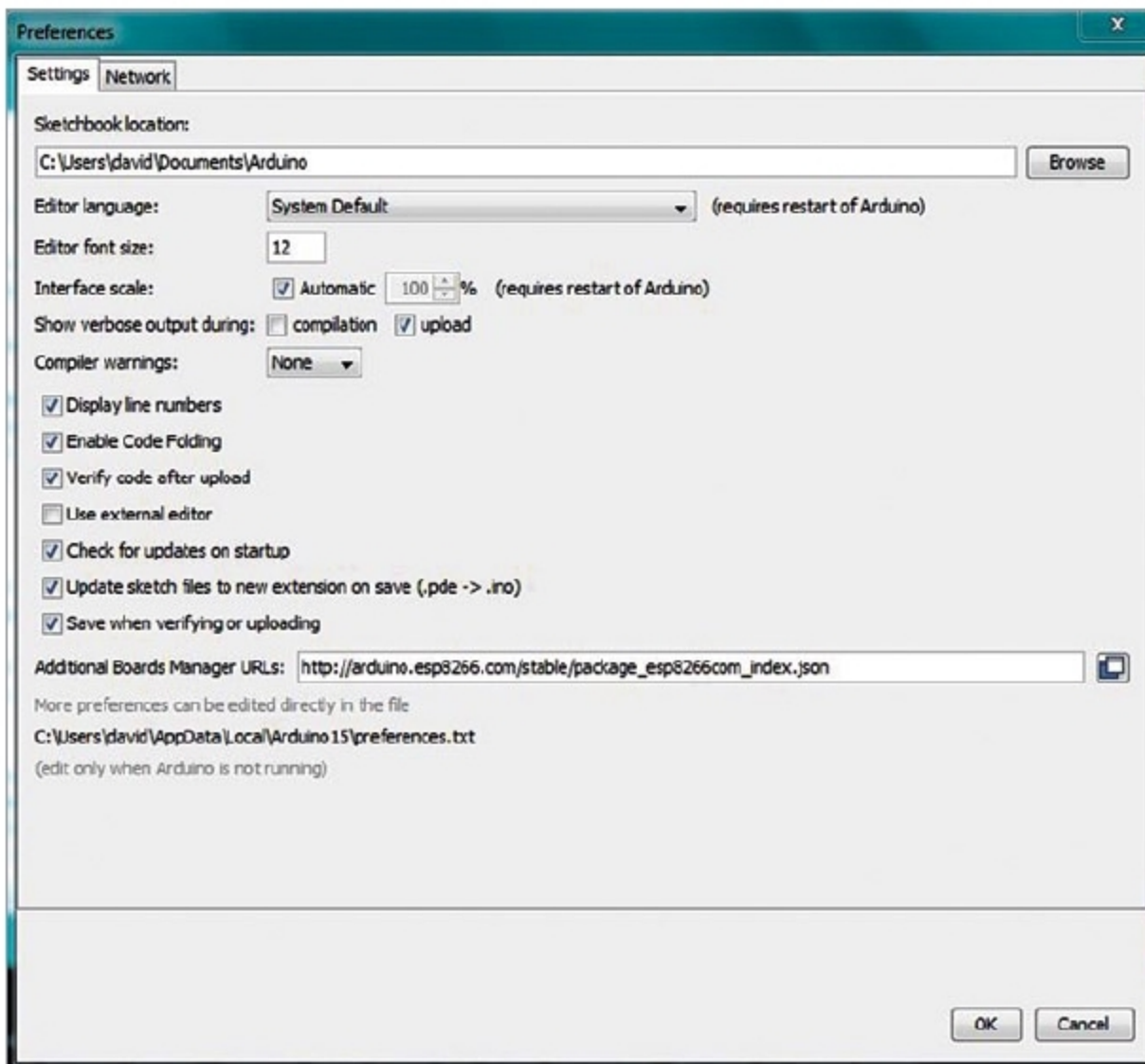


Fig. 2: Additional Board Manager URLs



Fig. 3: Boards Manager window

the computer, go to Tools→Port and select the correct COM port.

Arduino program

The program (device_web.ino) is written in Arduino programming language and compiled using Arduino IDE. Open the program into Arduino IDE and make necessary changes to the program (as explained below). Save and then upload the program to NodeMCU.

The following lines can be modified in Arduino program/sketch according to requirements:

```
#define stationMode false // if
false, ESP8266 is in SAP mode or else
it is in STA mode. (STA mode is tested
at EFY Lab.)
```

```
const char* ssid = "*****" //
```

string is used to store the access point name to ESP8266 module. It will connect to SAP, which ESP8266 will broadcast.

```
const char* password =
"*****" // string is used to store
the password of the access point or
SAP.
```

```
String username = "admin" //
string is used to store the username of
the login page.
```

```
String loginPassword = "admin"
// string is used to store the password
of the login page.
```

```
String headingMessage = " " //
string is used to set the headlines of
the webpage.
```

```
int noOfTry = 10 // 10 is the
number of false entries allowed.
```

```
int lockDownTime = 180000 //
180000 is lockdown time (in millisec-
onds).
```

```
int cookieExpireTime = 60000 //
60000 is cookie expiration time (in
milliseconds).
```

```
int refreshTime = 10 // 10 is
refresh time (in seconds) for status
check webpage.
```

Some functions used in this program are explained below.

statuscheck() function is called when a URL is requested from the browser, for example, IPADDRESS/status URL.

generateCookie() is used to generate a 32-bit cookie by taking characters randomly from string ranchars.

if_authenticated() is used to check whether the cookie is valid or not.

StatusCheck() is used to check the status of output pins.

background-color: #ff1d00 value can be changed to change the colour of buttons under void webpage().

Construction and testing

An actual-size PCB layout of the device control is shown in Fig. 4 and its components layout in Fig. 5. Power up NodeMCU board. If NodeMCU is configured as SAP mode, then connect it to the access point broadcast by NodeMCU. Open the Web browser on a cellphone or computer. Type 192.168.4.1 (default IP) in the address bar and press Enter.

If NodeMCU is configured as STA mode, use the IP address of NodeMCU. To find out the IP address of NodeMCU, you may refer to <https://tttapa.github.io/ESP8266/Chap07%20-%20Wi-Fi%20Connections.html>

Open the Web browser on a cellphone or computer. Type the IP address of NodeMCU in the address bar and press Enter.

The login webpage will look as shown in Fig. 6. IP address of NodeMCU may be different in your case. Note that, if there is no activity for some time on the login page, it will automatically log out after the session expires. If an invalid page is requested, it will give an error message, as shown in Fig. 7.

Click on Check Status shown in Fig. 6 to see the status of all four devices. The webpage will refresh as per